Sketch Design Options — Candidates for User Study

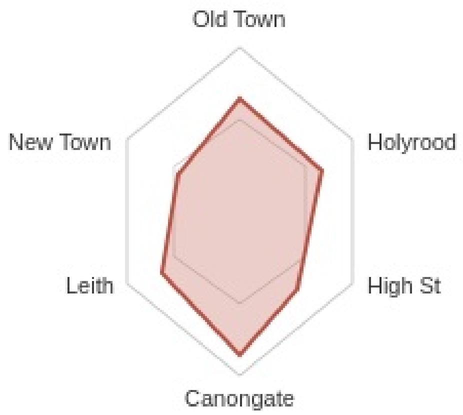
A Visual Trace Map of Edinburgh Place Names in Literature | Merritt Wang | for discussion with Uta Hinrichs

Direction 1: Author Spatial Fingerprint

Success criterion: Can users identify an author from the shape alone, on a one-to-one basis — i.e. different authors should not produce visually similar shapes (one-to-one, not many-to-one).

① Radar Chart

General



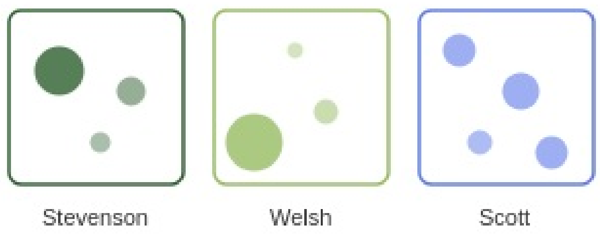
A fixed set of frequently-mentioned places becomes the axes; an author's mention frequency for each place sets the point on that axis, forming a polygon.

Strength Clear structure — the direction the polygon bulges towards directly reflects the author's spatial preference

Risk Discriminability depends heavily on which axes are chosen; a poor choice could make all authors look similar

② Small Multiples

General / Large samples



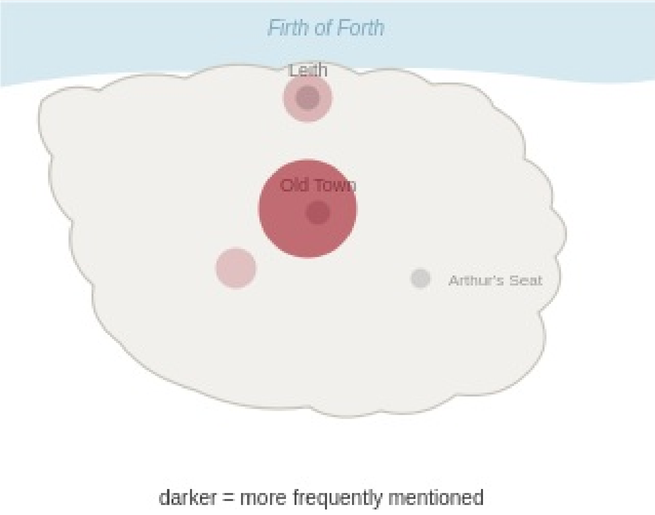
Each author gets a small individual panel, using bubble size/position to show their place-name distribution; all panels are shown side by side.

Strength Scales well — suitable for comparing 5-10 authors at once without lines crossing into clutter

Risk Each individual panel carries less information on its own; meaning depends on side-by-side comparison

③ Geographic Density Map

General public



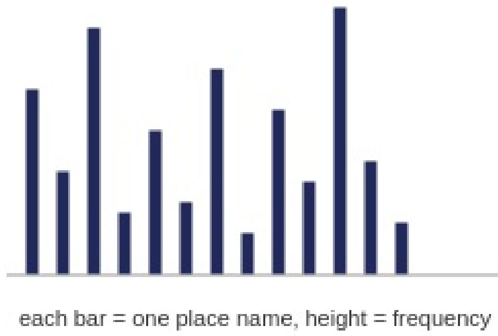
Uses a simplified version of Edinburgh's actual coastline and key landmarks as the base map; areas an author mentions more frequently are shaded darker.

Strength Grounded in real geography rather than an abstract shape — more intuitive for audiences with no visualisation background

Risk If an author's mentions concentrate heavily around the city centre, discriminability between authors may be limited

④ Bar-code Fingerprint

Domain experts



Each place name becomes a vertical bar; bar height encodes mention frequency. All bars are arranged in a fixed order, forming a bar-code-like silhouette.

Strength Minimal and information-dense; well suited to displaying many authors side by side as a 'fingerprint wall'

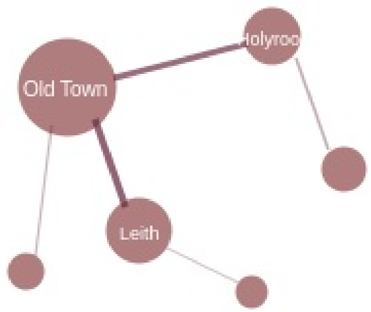
Risk More abstract than a radar chart for users unfamiliar with data charts

Direction 2: Narrative Topology Network

Success criterion: Can users perceive that "narrative proximity" and "geographic proximity" are two different things — i.e. the legibility of the relational structure itself.

① Force-directed Network

Domain experts



size = mention freq, edge = co-occurrence strength

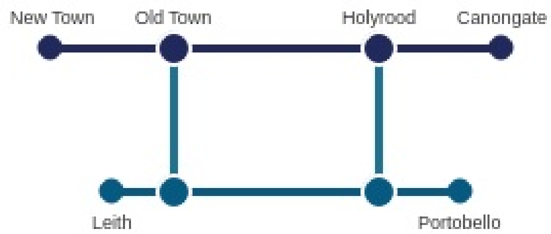
Places are nodes (size = mention frequency); co-occurrence relationships are edges (thickness = co-occurrence strength), automatically clustered by a layout algorithm.

Strength Naturally reveals clustering structure between places, surfacing unexpected groupings

Risk Becomes a visual 'hairball' once node count grows; harder for general audiences to read

② Metro-map Topology

General public



connections only — geography discarded

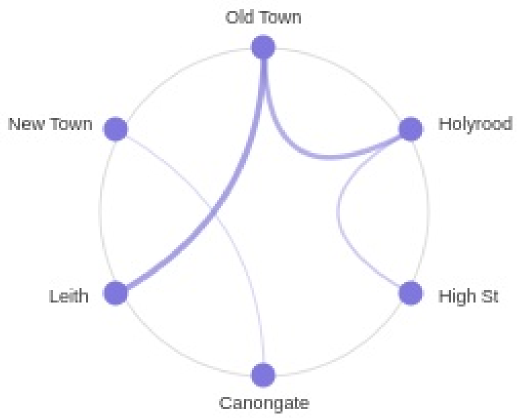
Addresses the same co-occurrence question, but forces a strict horizontal/vertical 'subway line' style, echoing the London Tube Map referenced in the project brief.

Strength Strong sense of visual order; narrative relationships are translated into a clear, readable 'interchange' logic

Risk Schematic metro-style layout is a known hard problem in graph drawing once place count grows — may need to limit the number shown

③ Chord Diagram

General



arc thickness = co-occurrence strength

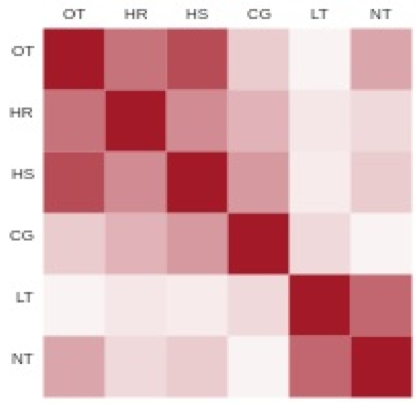
Places are arranged around a circle; co-occurrence between two places is drawn as an arc connecting them, with arc thickness indicating strength.

Strength Strong global view — easy to spot 'hub' places where many connections converge

Risk Becomes crowded once arcs multiply; weak long-distance relationships can get lost

④ Adjacency Matrix Heatmap

Domain experts



cell darkness = co-occurrence strength

Both axes list all places; cell darkness encodes the co-occurrence strength between the row and column place — the most 'precise' representation.

Strength Most accurate numerical relationships, with no ambiguity introduced by visual layout — suited to close analysis

Risk Unfriendly to general audiences; visually resembles a data table rather than a story

Direction	Candidate design	Primary discrimination cue	Best suited for
Fingerprint	Radar chart	Polygon shape	General
	Small multiples	Side-by-side scanning	General / large samples
	Geographic density map	Position of colour blocks	General public
	Bar-code fingerprint	Silhouette contour	Domain experts
Topology	Force-directed network	Natural clustering	Domain experts
	Metro-map topology	Ordered, legible connections	General public
	Chord diagram	Hub identification	General
	Adjacency matrix heatmap	Precise numerical relationships	Domain experts

Study design note: Plan to run an online study with two participant groups — domain professionals (geography/literature) and a broader general audience. Sketches will be introduced by explaining the project's purpose only, without over-explaining what each chart means, so participants can independently attempt the task of distinguishing authors or identifying narrative relationships. A simple vote will also be collected on which design feels most intuitive.